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DESIGN OF EXPERIMENTS IN NON-LINEAR SITUATIONS

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1. Outline of the problem

We suppose that some response η is a known function

$$\eta = f(\xi_1, \xi_2, \ldots, \xi_k;\ \theta_1, \theta_2, \ldots, \theta_p) = f(\boldsymbol{\xi};\ \boldsymbol{\theta}) \tag{1}$$

of k variables whose levels are denoted by the elements $\xi_1, \ldots, \xi_i, \ldots, \xi_k$ of the vector $\boldsymbol{\xi}$ and of p parameters $\theta_1, \ldots, \theta_r, \ldots, \theta_p$ elements of the vector $\boldsymbol{\theta}$, and that this function is not necessarily linear in either the variables or the parameters. We further suppose that it is possible to perform a series of experiments or trials in each one of which the response is observed at a particular chosen set of levels of the variables decided in advance.

The problem here considered is that of selecting a programme of trials such that they may be expected to provide results from which the p parameters can be estimated with high accuracy. In general the experimental programme may be defined by an $N \times k$ matrix $\mathbf{D} = \{\xi_{iu}\}$ called the design matrix. The uth row $\boldsymbol{\xi}'_u$ of this matrix with elements $\xi_{1u}, \ldots, \xi_{iu}, \ldots, \xi_{ku}$ provides the levels of the k variables at which the response is to be observed in the uth trial.

1·1. *Example*

The following example illustrates the situation we have in mind. Suppose a consecutive chemical reaction is under study in which a substance A decomposes to form substance B which then in turn decomposes to form substance C. With the assumptions that the reactions are first order and irreversible it may be shown that, after time ξ_1 has elapsed, the yield η of intermediate product B is given by

$$\eta = \frac{\theta_1}{\theta_1 - \theta_2}\{\exp(-\theta_2 \xi_1) - \exp(-\theta_1 \xi_1)\}, \tag{2}$$

where θ_1 and θ_2 are constants measuring the specific rates of the first and second decompositions, respectively.

The problem is to choose a set of times ξ_{1u}, $u = 1, 2, \ldots, N$, at which to observe the yield so that from these observations θ_1 and θ_2 can be estimated as accurately as possible, allowing for experimental error in the observations. In this example the design matrix $\mathbf{D}$ would simply consist of a single column whose N elements were the times $\xi_{11}, \ldots, \xi_{1u}, \ldots, \xi_{1N}$ at which the yield was to be observed and $\boldsymbol{\theta}$ would contain the two elements θ_1 and θ_2. More generally there will be k variables (for example, ξ_1 = time, ξ_2 = temperature, ξ_3 = concentration, etc.) and the design matrix will contain k columns.

1·2. *Practical limits to levels of variables*

In practice, levels at which variables such as time, temperature, concentration and dosage can be set are usually restricted. For example, in the study of the chemical reaction above, the feasible range for the variable time (ξ_1) will be from zero to some value ξ_1(max) which will be the longest time which could be contemplated in an actual experiment. In general, there

will exist some experimental region R in the $\boldsymbol{\xi}$ space which delimits the area within which trials can actually be conducted. In a chemical reaction, for example, such limitation may occur from explosive hazards associated with certain combinations of temperature and pressure. In other cases it may be simply that it is experimentally inconvenient to conduct trials at certain combinations of conditions. In some cases the experimental region can be defined by a series of inequalities in the ξ's of the type

$$\xi_i(\min) \leqslant \xi_i \leqslant \xi_i(\max) \quad (i = 1, 2, \ldots, k).$$

In general, however, the definition of R may be more complicated.

1·3. *Criterion for selection of design*

We denote the response observed at the uth set of experimental conditions by y_u and we suppose that

$$E(y_u) = \eta_u = f(\boldsymbol{\xi}_u, \boldsymbol{\theta}), \tag{3}$$

and

$$E(y_u - \eta_u)(y_v - \eta_v) = \begin{cases} \sigma^2 & u = v \\ 0 & u \neq v \end{cases} \quad (u, v = 1, 2, \ldots, N), \tag{4}$$

where, in general, σ^2 is unknown. The true values of the parameters are denoted by $\theta_{10}, \theta_{20}, \ldots, \theta_{p0}$, the elements of the vector $\boldsymbol{\theta}_0$. The partial derivatives of the response function with respect to the rth parameter θ_r for the uth set of experimental conditions, taken at the point $\boldsymbol{\theta}_0$, is denoted by

$$f_{ru} = \left[\frac{\partial f(\boldsymbol{\xi}, \boldsymbol{\theta})}{\partial \theta_r}\right]_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}. \tag{5}$$

Finally, the $N \times p$ matrix of these derivatives is

$$\mathbf{F} = \{f_{ru}\}. \tag{6}$$

It is well known that the least squares estimates $\hat{\boldsymbol{\theta}}$ obtained by minimizing the sum of squares

$$\sum_{u=1}^{N} \{y_u - f(\boldsymbol{\xi}_u, \boldsymbol{\theta})\}^2 \tag{7}$$

and given by solving the normal equations

$$\sum_{u=1}^{N} \{y_u - f(\boldsymbol{\xi}_u, \hat{\boldsymbol{\theta}})\} \{\partial f(\boldsymbol{\xi}_u, \boldsymbol{\theta})/\partial \theta_r\}_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} = 0 \quad (r = 1, 2, \ldots, p) \tag{8}$$

have a variance covariance matrix which is approximated by $(\mathbf{F}'\,\mathbf{F})^{-1}\sigma^2$.

We shall proceed by attempting to choose $\mathbf{D}$ so that the determinant $|(\mathbf{F}'\mathbf{F})^{-1}|$ is made as small as possible. If we assume that each $f(\boldsymbol{\xi}_u, \boldsymbol{\theta})$ is approximately linear in the neighbourhood of $\boldsymbol{\theta}_0$ we are thus minimizing Wilk's generalized variance for the estimates. The logic of this choice, also made by Wald (1943), can be seen by noting that if the estimates are approximately normally distributed, the determinant $|(\mathbf{F}'\,\mathbf{F})^{-1}|$ is proportional to the volume contained within any specific ellipsoidal probability contour for $\boldsymbol{\theta}$ about $\boldsymbol{\theta}_0$ in the space of the parameters. The criterion chosen ensures that any such probability contour includes the smallest volume.

To understand the significance of this criterion in general non-linear situations it seems desirable to consider the relationship between sample space, the 'solution locus' within sample space, and parameter space. The point $\boldsymbol{\theta}$ in parameter space can be mapped on to the point $P(\boldsymbol{\theta})$ in sample space with co-ordinates $\eta_u = f(\boldsymbol{\xi}_u, \boldsymbol{\theta})$ $(u = 1, \ldots, N)$. We define the part of sample space covered by this mapping as the 'solution locus'.

Now, given any point $P(\boldsymbol{\theta})$ on the solution locus, we can make an orthogonal transformation $\boldsymbol{\eta} = \mathbf{H}\boldsymbol{\zeta}$ of co-ordinates in sample space such that the solution locus in the immediate neighbourhood of $P(\boldsymbol{\theta})$ coincides with part of the p-dimensional hyperplane $\zeta_i =$ constant $(i = p+1, \ldots, N)$. We can then interpret $|(\mathbf{F}'\mathbf{F})^{-1}|$ as the square of the Jacobian of the transformation from co-ordinates $(\zeta_1, \ldots, \zeta_p)$ in sample space to $(\theta_1, \ldots, \theta_p)$ in parameter space, when $P(\boldsymbol{\theta}) = P(\boldsymbol{\theta}_0)$. So the criterion that $|(\mathbf{F}'\mathbf{F})^{-1}|$ is to be made as small as possible means geometrically that the p-dimensional volume of that part of parameter space associated with a region of prescribed small volume round $P(\boldsymbol{\theta}_0)$ on the solution locus shall be as small as possible. Since we do not know $\boldsymbol{\theta}_0$ in advance, we must assume that we can find a design that makes this Jacobian about as small as possible for all points $P(\boldsymbol{\theta})$ near $P(\boldsymbol{\theta}_0)$.

Let us now assume that the part of the solution locus near $P(\boldsymbol{\theta}_0)$ can be approximated by a p-dimensional linear subspace of sample space. If we also assume that the experimental error distributions are normally distributed as well as satisfying (3) and (4), then the confidence region for $\boldsymbol{\theta}$ based on the likelihood ratio criterion consists of all points associated with points on the solution locus within a p-dimensional sphere centred on the least squares estimate $P(\hat{\boldsymbol{\theta}})$, and with radius independent of the experimental design. Any design that gives an adequate approximation to the smallest possible value of the Jacobian for all points round $P(\boldsymbol{\theta}_0)$ therefore approximately minimizes the p-dimensional volume of the confidence region in parameter space.

1·4. *Necessity for preliminary estimates*

We see that the efficiency of different possible designs depends upon the matrix $\mathbf{F}$ whose elements are the values of the derivatives of the response function with respect to the θ's at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. Now the derivatives $\partial f/\partial\theta_r$ can only be independent of the values actually taken by the parameter θ_r if the response function f is linear in θ_r. For non-linear response functions the values of the derivatives, and hence the efficiency of any particular design, depend upon the actual values of the p-parameters. If we are to suppose that effective design is possible at all, we must also assume therefore that *something* is known about the values of the parameters in advance.

In practical problems it will almost invariably be the case that some such information is available, and this will then provide the basis of a first design. As is inevitably the case here and elsewhere, the real effectiveness of the design will depend upon the reliability of the information on which it is based. In this paper, therefore, we assume that preliminary values $\boldsymbol{\theta}^*$ of the parameters are available and will proceed so that the design would be optimal if these quantities were the true values. Some further discussion of the general situation for which this theory is appropriate is given by Box (1957).

1·5. *Possible extensions of the problem*

It is clear that in some examples normality of distribution and constancy of variance would not be possible at all points on the response function if only because the variation might be limited either upwards or downwards. Two obvious extensions, therefore, of this work would be to generalize to a maximum likelihood solution in which any distribution is considered or to consider weighted least squares. Valuable work in this direction has already been carried out by Elfving (1952) and Chernoff (1953). Again, in some cases we might wish to make a transformation of the model before the present theory was applied.

Apart from questions of appropriate distribution assumptions it should be possible to extend this work in at least two other directions. In the first place in different problems the reliability of the preliminary values $\boldsymbol{\theta}^*$ will be different, and we might take account of this by using some rough estimate of the prior probability distribution. Again, some observations might be cheaper than others. One could then attempt to find the design giving the most information for a given cost. Finally, it would be desirable in many circumstances to use a sequential procedure in which estimates from a preliminary experiment would be used to plan a second experiment, the results of which would in turn be used to plan a third experiment, and so on. These possibilities will not be further discussed here but we hope it may be possible to pursue them in later work.

2. Solution

Suppose that we are given the functional form

$$\eta_u = f(\boldsymbol{\xi}_u, \boldsymbol{\theta}) \tag{9}$$

and a set of preliminary values $\boldsymbol{\theta}^*$. Then we shall choose the design $\mathbf{D}$ such that $|\mathbf{F}^{*\prime}\mathbf{F}^*|$ is a maximum within the experimental region R in the $\boldsymbol{\xi}$-space, where

$$\mathbf{F}^* = \{f_{ru}^*\} \tag{10}$$

and

$$f_{ru}^* = \left[\frac{\partial f(\boldsymbol{\xi}_u, \boldsymbol{\theta})}{\partial \theta_r}\right]_{\boldsymbol{\theta}=\boldsymbol{\theta}^*}. \tag{11}$$

To limit to manageable proportions the possibilities we need to consider, we will in this paper assume that the number of trials (that is the number of combinations of levels of the variables) is equal to the number p of parameters to be estimated. Each of the p trials could of course be replicated to attain any desired degree of precision, but it should be noted that an r-fold replication of a design that minimizes the generalized variance using p observations, does not necessarily minimize the generalized variance for all possible experiments involving rp observations. Experiments in which the number of parameters estimated is equal to the number of trials are not perhaps intuitively appealing. This is (i) because one is usually not completely assured that the model is correct; so that one ordinarily has in fact a twofold objective: to check the model and to estimate the parameters in it if it proves to be representationally adequate, and (ii) because designs in which there are more than the minimum number of points may provide greater protection against gross errors in guessing $\boldsymbol{\theta}^*$.

In order for a design to be efficient in 'checking' the model, we must be specific about what sorts of departure from the model we suspect. In so far as this can be done, we should be able to write an 'alternative' wider model which makes allowance for these possible departures *in terms of extra parameters to be estimated.* In this way, within the framework discussed we can achieve both objectives mentioned above. As a simple illustration consider the chemical reaction discussed earlier. Here A was decomposed to form B which in turn decomposed to C. In a particular problem the second decomposition might be unlikely, in which case the primary model in mind for the production of B would be

$$\eta = 1 - e^{-\theta_1 \xi}. \tag{12}$$

However, in the light of possible decomposition of B to C it would be appropriate to plan the experiment so as to be effective in estimating both θ_1 and θ_2 in the model of equation (2).

When nothing is known about the type of departures which may occur we are confronted once more with the fact already mentioned that the effectiveness of a design must depend on

the reliability of information on which it is based. If this information is non-existent then we have no basis on which to plan. In practice, however, the situation may arise where some information is available which is difficult to express in terms of extra parameters in the model, but is manifested in feelings about the way the response surface may be distorted. In this case further points might be added to the design in an intuitive way, so as to tie down doubtful areas of the surface, while at the same time bearing in mind the locations already determined which ensure maximum reliability for the principal parameters when the model is adequate.

When gross errors may occur in guessing $\boldsymbol{\theta}^*$ the present theory can be used to explore the changes that would occur in the position of the optimal p design points with changes in $\boldsymbol{\theta}^*$ over the region of uncertainty and the losses in efficiency which occurred with given errors in $\boldsymbol{\theta}^*$. If more than p-points were available these might then be chosen so as to minimize the loss of efficiency as far as possible.

2·2. *A geometrical formulation*

For that class of designs for which the number of trials is equal to the number of parameters, $\mathbf{F}^*$ is a square $(p \times p)$ matrix. Consequently,

$$|\mathbf{F}^{*\prime}\mathbf{F}^*| = |\mathbf{F}^*|^2 \tag{13}$$

and we should choose the design $\mathbf{D}$ so that the modulus of $|\mathbf{F}^*|$ is a maximum in the experimental region R.

In understanding the problem to be solved and in showing the properties of solutions in simple cases, the following geometrical formulation has been found of help. Consider a p-dimensional space in which the co-ordinates are the derivatives

$$\left(\frac{\partial f(\boldsymbol{\xi}, \boldsymbol{\theta})}{\partial \theta_r}\right)_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} = \phi_r(\boldsymbol{\xi}) \quad (r = 1, 2, \ldots, p), \tag{14}$$

where, it will be noted, $\phi_r(\boldsymbol{\xi})$ is a function of $\boldsymbol{\xi}$ only. The surface in the space of the derivatives generated by all permissible values of $\boldsymbol{\xi}$ we call the 'design locus'. Thus with any particular choice of p 'design points', $\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \ldots, \boldsymbol{\xi}_u, \ldots, \boldsymbol{\xi}_p$, in the ξ-space for a particular design matrix $\mathbf{D}$, there will be corresponding points on the design locus in the space of the derivatives $\phi_r(\boldsymbol{\xi})$.

Now the volume of a simplex in p-space formed by p-points

$$(z_{11}, z_{12}, \ldots, z_{1p}); \ldots; (z_{s1}, z_{s2}, \ldots, z_{sp}); \ldots; (z_{p1}, z_{p2}, \ldots, z_{pp})$$

and the origin $(0, 0, \ldots, 0)$ is proportional to the determinant of the $p \times p$ matrix $\{z_{sr}\}$. Consequently, we must find p design points $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_u, \ldots, \boldsymbol{\xi}_p$ for which the corresponding points in the space of the derivatives, with co-ordinates defined by equation (14), form with the origin the simplex of greatest volume.

2·3. *Example*

As a simple illustration, consider the example given in § 1·1 of a consecutive first-order chemical reaction. Suppose that we are given preliminary guesses θ_1^*, θ_2^* for the values of the parameters. We wish to choose two values ξ_{11}, ξ_{12} of time so that the best estimates will be available for θ_1, θ_2. We have

$$\eta = f(\xi_1; \theta_1, \theta_2) = \frac{\theta_1}{\theta_1 - \theta_2}\{\exp(-\theta_2 \xi_1) - \exp(-\theta_1 \xi_1)\}. \tag{15}$$

The region R is defined by $$0 \leqslant \xi_{11} \leqslant \xi_{12} \leqslant \xi_1(\max). \tag{16}$$

but for the moment no upper limit $\xi_1(\max)$ will be imposed. We have to choose ξ_{11}, ξ_{12} so as to maximize the modulus of the determinant

$$\Delta = |\mathbf{F}^*| = \begin{vmatrix} \phi_1(\xi_{11}) & \phi_2(\xi_{11}) \\ \phi_1(\xi_{12}) & \phi_2(\xi_{12}) \end{vmatrix}. \tag{17}$$

If the preliminary guesses are

$$\theta_1^* = 0{\cdot}7, \quad \theta_2^* = 0{\cdot}2, \tag{18}$$

then

$$\phi_1(\xi_1) = (0{\cdot}8 + 1{\cdot}4\xi_1)\exp(-0{\cdot}7\xi_1) - 0{\cdot}8\exp(-0{\cdot}2\xi_1), \tag{19}$$

$$\phi_2(\xi_1) = (2{\cdot}8 - 1{\cdot}4\xi_1)\exp(-0{\cdot}2\xi_1) - 2{\cdot}8\exp(-0{\cdot}7\xi_1). \tag{20}$$

We can of course substitute equations (19) and (20) in (17) and calculate the values ξ_{11}, ξ_{12} which maximize the modulus of Δ. It is instructive, however, to study this solution in some detail using the geometrical construction. The dotted line in Fig. 1 is a graph of the function

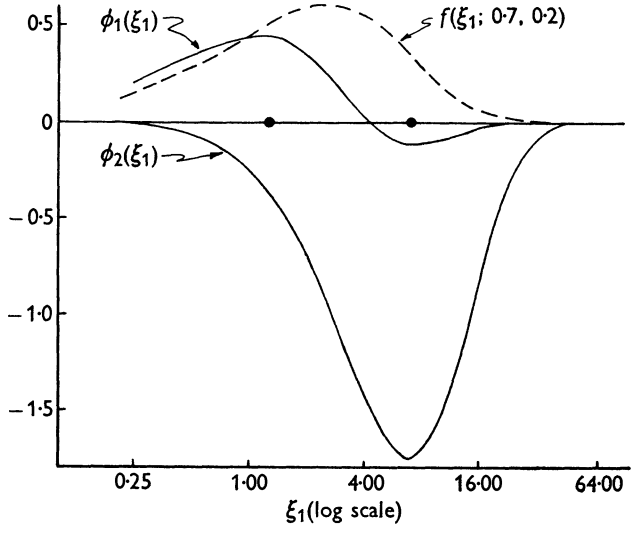

Fig. 1. The function $f(\xi_1;\ 0{\cdot}7,\ 0{\cdot}2)$ and its derivatives. The optimal solution for the design is indicated by the heavy dots.

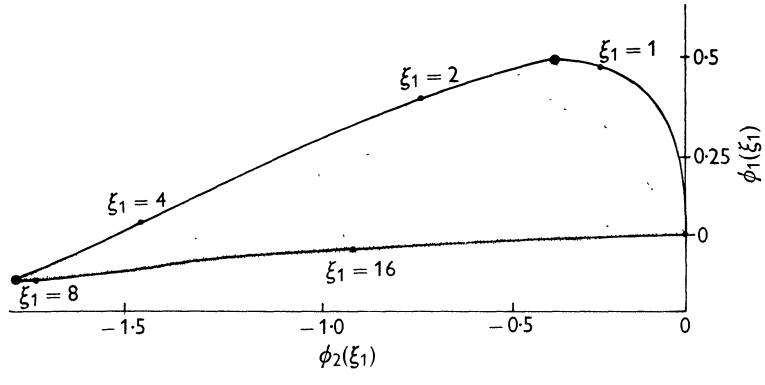

Fig. 2. The design locus for the function $f(\xi_1;\ 0{\cdot}7,\ 0{\cdot}2)$ with the optimal solution indicated by heavy dots.

$f(\xi_1;\ 0{\cdot}7, 0{\cdot}2)$. On the same diagram are shown the values of the derivatives $\phi_1(\xi_1)$, $\phi_2(\xi_1)$. Fig. 2 shows $\phi_1(\xi_1)$ plotted against $\phi_2(\xi_1)$ for values of ξ_1 covering the range of interest. This is the curve we call the design locus.

In terms of Fig. 2 the problem is to find those two points ξ_{11}, ξ_{12} on this locus, which together with the origin form the vertices of the triangle of greatest area. The exact solution obtained by numerical methods described later is $\xi_{11} = 1{\cdot}23$, $\xi_{12} = 6{\cdot}86$, corresponding to the triangle denoted by the shaded area in Fig. 2. This solution is indicated in Fig. 1 by the points at $\xi_{11} = 1{\cdot}23$ and $\xi_{12} = 6{\cdot}86$. From the graph of the function $f(\xi_1;\ 0{\cdot}7, 0{\cdot}2)$ it will be seen that for this particular example the values are chosen so as to lie on either side of the maximum at points of moderately high yield. The graphs of the derivatives illustrate how the criterion seeks points at which the derivatives are large in absolute value, but also, so far as possible, uncorrelated.

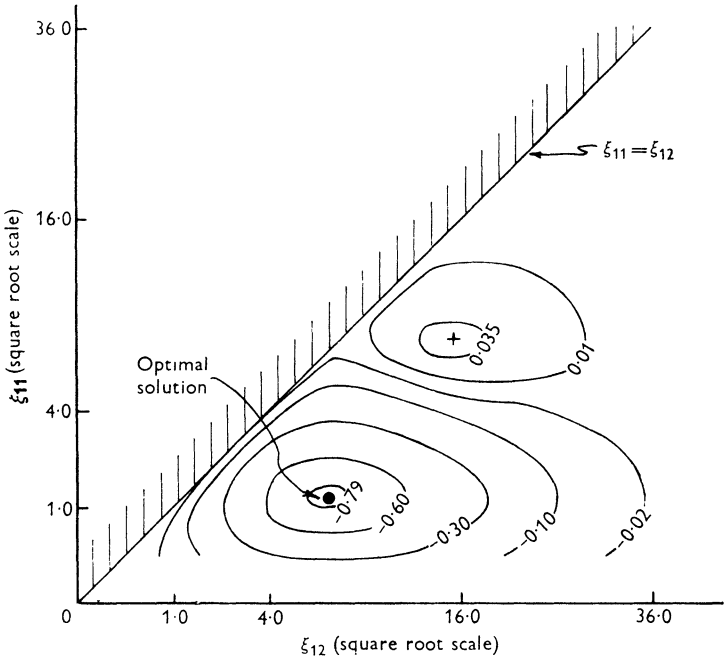

Fig. 3. Contours of Δ for the function $f(\xi_1;\ 0{\cdot}7,\ 0{\cdot}2)$ in the space of ξ_{11}, ξ_{12} for $0 \leqslant \xi_{11} \leqslant \xi_{12}$.

2·4. *Some problems in determining the solution*

This example may be used to illustrate some of the problems in determining the solution. Fig. 3 shows contours of Δ in the space ξ_{11}, ξ_{12} as defined by equation (17) within the region $0 \leqslant \xi_{11} \leqslant \xi_{12}$. It is seen that this surface contains *two* local extrema, both corresponding to local maxima in the *modulus* of the determinant. In terms of the actual value of the determinant, there is a minimum of $\Delta = -0{\cdot}810$ at the values $\xi_{11} = 1{\cdot}23$, $\xi_{12} = 6{\cdot}86$ and a maximum of $\Delta = 0{\cdot}039$ at about $\xi_{11} = 7{\cdot}5$, $\xi_{12} = 15{\cdot}0$. The latter value is proportional to the area of a small triangle that can be fitted into the concavity on the lower side of the design locus in Fig. 2. This solution is of no practical interest, since the absolute value of the determinant is only about one-twentieth of the value which corresponds to the other solution. However, it is of some importance to note that surfaces with multiple extrema can occur, and may complicate the problem of finding best solutions in more complex cases.

2·5. *Numerical solution*

Although in particular examples there is an analytic solution to the problem of maximizing the modulus of the determinant $|\mathbf{F}^*|$, often numerical methods must be employed. Such methods are attractive because they allow us to obtain solutions even when the function $f(\boldsymbol{\xi}, \boldsymbol{\theta})$ is not given explicitly. In chemical examples, for instance, a kinetic mechanism can often be postulated in terms of a series of differential equations which may have no explicit solution. The equations may be integrated numerically, however, on a computer and the quantities $\phi_r(\boldsymbol{\xi})$ for any trial value $\boldsymbol{\xi}$ obtained from difference formulae. Iterative methods may then be employed to find the maximum value of the modulus of the determinant.

In the particular example studied it is seen from Fig. 3 that in the neighbourhood of the optimum solution the surface is well conditioned, containing no oblique ridges, and that consequently a numerical method in which the variables were adjusted one at a time would be successful in determining the maximum. It is fairly easy to see that the satisfactory state of conditioning indicated by the 'symmetrical' maxima found in this problem cannot be expected to be general. Often the Δ surface would contain attenuated ridges which in the later stages of iteration would give rise to difficulties in numerical solution when the method of adjusting 'one variable at a time' or 'steepest ascent' procedures were used. A method which can be employed when the final solution is approached is the second-order procedure similar to that employed by Koshal (1933) in solving maximum likelihood equations. This consists of fitting a second-degree polynomial to values of the determinant calculated at suitably arranged series of points.

Calculations for improved values of ξ_{11} and ξ_{12}, starting from the values $\xi_{11} = 1{\cdot}10$, $\xi_{12} = 6{\cdot}50$ (obtained from Fig. 2) are shown for the present example in Table 1.

Table 1. *Calculation of improved values of maximum*

ξ_{11}	ξ_{12}	$-\Delta$
1·10	6·50	0·803 336
1·20	6·50	0·807 992
1·30	6·50	0·806 709
1·10	7·00	0·804 758
1·10	7·50	0·798 920
1·20	7·00	0·809 632

Writing $x_1 = 5(\xi_{11} - 1{\cdot}10)$, $x_2 = (\xi_{12} - 6{\cdot}50)$, the second-degree interpolation polynomial is

$$-\Delta = 0{\cdot}803336 + 0{\cdot}015251x_1 + 0{\cdot}010104x_2 - 0{\cdot}011878x_1^2 - 0{\cdot}014250x_2^2 + 0{\cdot}000872x_1x_2$$

which has a maximum at $\quad x_1 = 0{\cdot}66, \quad x_2 = 0{\cdot}37,$

that is $\quad \xi_{11} = 1{\cdot}23, \quad \xi_{12} = 6{\cdot}87.$

A second iteration provided values in substantial agreement with these, namely, $\xi_{11} = 1{\cdot}23$ and $\xi_{12} = 6{\cdot}86$. It is clear that because of difficulties of possible multiple maxima and the complexities of the Δ surface, considerable caution is necessary in applying numerical

methods to problems in which there are a large number of parameters. It is hoped to discuss such cases in a later paper. The examples which follow contain only a few parameters but nevertheless are thought to be of some inherent interest.

3. Examples of non-linear designs

3·1. *Simple decay and growth law*

Consider the model

$$\eta = \theta_1 \exp(\theta_2 \xi_1), \quad 0 \leqslant \xi_1(\min) \leqslant \xi_{11} < \xi_{12} \leqslant \xi_1(\max). \tag{21}$$

With ξ representing time and with $\theta_1 > 0$, $\theta_2 < 0$, this is the well-known first-order decay law. With $\theta_1 > 0, \theta_2 > 0$, the function can represent growth phenomena. In application both lower and upper limits $\xi_1(\min)$ and $\xi_1(\max)$ may be imposed corresponding to the earliest and latest points in time at which observations can be made. The value of the determinant is

$$\Delta = \theta_1 \exp\{\theta_2(\xi_{11}+\xi_{12})\}(\xi_{12}-\xi_{11}). \tag{22}$$

The modulus of which for $\theta_2^* < 0$ attains its maximum value when

$$\left.\begin{aligned} \xi_{11} &= \xi_1(\min), \\ \xi_{12} &= \xi_1(\min) - 1/\theta_2^*. \end{aligned}\right\} \tag{23}$$

Substituting these values into equation (21) we see that the best design will be that in which a proportion $e^{-1} = 36{\cdot}8\,\%$ of the amount of substance present at the first observation remains at the second observation.

For $\theta_2^* > 0$ the modulus of Δ is a maximum when

$$\left.\begin{aligned} \xi_{11} &= \xi_1(\max) - 1/\theta_2^*, \\ \xi_{12} &= \xi_1(\max). \end{aligned}\right\} \tag{24}$$

The first observation should thus be made at a time when $36{\cdot}8\,\%$ of the growth expected at the final observation has occurred.

If instead of the amount of substance decayed we consider the amount of product formed from a simple decay law, we have

$$\eta = \theta_1\{1-\exp(\theta_2\xi_1)\}$$

for

$$0 \leqslant \xi_1(\min) \leqslant \xi_{11} < \xi_{12} \leqslant \xi_1(\max). \tag{25}$$

The value of the determinant is

$$\Delta = \theta_1[\{1-\exp(\theta_2\xi_{12})\}\xi_{11}\exp(\theta_2\xi_{11}) - \{1-\exp(\theta_2\xi_{11})\}\xi_{12}\exp(\theta_2\xi_{12})], \tag{26}$$

the modulus of which has a maximum when

$$\xi_{12} = \xi_1(\max), \tag{27}$$

$$\xi_{11} = -\frac{1}{\theta_2^*} - \frac{\xi_{12}\exp(\theta_2^*\xi_{12})}{1-\exp(\theta_2^*\xi_{12})}. \tag{28}$$

Substituting these values into equation (25) and letting γ_2 be the fraction by which the yield at time $\xi_{12} = \xi_1(\max)$ falls short of the maximum obtainable, we find that the optimum value of ξ_{11} corresponds to a yield falling short of the maximum by a fraction

$$\gamma_1 = \exp\{-[1+\gamma_2\log\gamma_2/(1-\gamma_2)]\}. \tag{29}$$

In particular if the second observation is taken after a sufficiently long time so that the asymptotic value is almost achieved, then γ_2 is close to zero and $\gamma_1 = 1/e$, so that the first observation should be made when the process is 63·2 % complete. As a second example, if the second observation is taken at a time when the process is 50 % complete, then $\gamma_1 = 0{\cdot}73$ and the first observation should be made at a time when the process is 27 % complete.

3·2. *Mitscherlich equation*

Consider the model

$$\eta = \theta_1 + \theta_2 \exp(\theta_3 \xi_1) \tag{30}$$

for

$$0 \leqslant \xi_1(\min) \leqslant \xi_{11} < \xi_{12} < \xi_{13} \leqslant \xi_1(\max).$$

With $\theta_3 < 0$ and $\theta_2 < 0$ this expression represents the well-known Mitcherlich law of diminishing returns. For example, in crop experiments with ξ_1 the amount of added fertilizer and η the amount of growth, $\theta_1 + \theta_2$ represents the growth when no fertilizer is added and θ_1 the hypothetical growth obtained from an infinite amount of fertilizer; θ_3 measures the rate at which the response to each additional increment of fertilizer decreases. The same model has been employed to relate the weight η of chickens to their cumulative feed consumption ξ_1 (Hendricks *et al.* 1932). With $\theta_3 > 0$ this model has been used to represent early growth (Brant, 1951).

The value of the determinant is

$$\begin{aligned}\Delta = \theta_2[&\xi_{11} \exp(\theta_3 \xi_{11}) \{\exp(\theta_3 \xi_{13}) - \exp(\theta_3 \xi_{12})\} + \xi_{12} \exp(\theta_3 \xi_{12}) \\ &\times \{\exp(\theta_3 \xi_{11}) - \exp(\theta_3 \xi_{13})\} + \xi_{13} \exp(\theta_3 \xi_{13}) \{\exp(\theta_3 \xi_{12}) - \exp(\theta_3 \xi_{11})\}].\end{aligned} \tag{31}$$

It is readily shown that irrespective of the signs of θ_1, θ_2, θ_3, the optimum design is obtained with

$$\xi_{11} = \xi_1(\min), \tag{32}$$

$$\xi_{12} = -\frac{1}{\theta_3^*} + \frac{\xi_{11} \exp(\theta_3^* \xi_{11}) - \xi_{13} \exp(\theta_3^* \xi_{13})}{\exp(\theta_3^* \xi_{11}) - \exp(\theta_3^* \xi_{13})}, \tag{33}$$

$$\xi_{13} = \xi_1(\max). \tag{34}$$

In particular in a fertilizer experiment if $\xi_{11} = \xi_1(\min)$ corresponded to no dressing of fertilizer and $\xi_{13} = \xi_1(\max)$ corresponded to a dressing for which the response fell short of the maximum by a certain fraction γ_2, then ξ_{12} would be a dressing for which the response fell short of the maximum possible by a fraction γ_1 where again γ_1 would be given by (29). When $\xi_1(\max)$ is sufficiently large to give an almost maximal response then the intermediate value of the design should be that expected to yield a response of 63·2 % of the maximum possible.

3·3. *Simple first-order decay with rate a function of temperature*

Consider the model

$$\eta = \exp(-\theta_1 \xi_1 e^{-\theta_2 \xi_2}) \tag{35}$$

for

$$0 \leqslant \xi_1(\min) \leqslant \xi_{11} < \xi_{12} \leqslant \xi_1(\max),$$
$$\xi_2(\min) \leqslant \xi_{21} < \xi_{22} \leqslant \xi_2(\max).$$

This would apply for example where η represented the fractional amount remaining at time $t = \xi_1$ of a material decaying according to the first-order law, when the rate was temperature-

sensitive. Assuming that the temperature effect follows the Arrhenius rule, θ_1 is the specific rate at some absolute temperature T_0, θ_2 is proportional to the 'activation energy' and $\xi_2 = T^{-1} - T_0^{-1}$ is the deviation of the reciprocal of the temperature, T, from the reciprocal of the base temperature T_0.

We find for preliminary values θ_1^*, θ_2^* that

$$\phi_1(\xi_1, \xi_2) = -\xi_1 \exp\{-(\theta_2^* \xi_2 + \theta_1^* \xi_1 e^{-\theta_2^* \xi_2})\} = \frac{\eta^* \ln \eta^*}{\theta_1^*},$$

$$\phi_2(\xi_1, \xi_2) = -\theta_1^* \xi_2 \phi_1(\xi_1, \xi_2) = -\xi_2 \eta^* \ln \eta^*, \tag{36}$$

where $\eta_u^* = \exp(-\theta_1^* \xi_{1u} e^{-\theta_2^* \xi_{2u}})$, the expected fraction of decaying substance remaining in the uth experiment. Thus

$$\operatorname{mod} \Delta = \operatorname{mod} \begin{vmatrix} \frac{\eta_1^*}{\theta_1^*} \ln \eta_1^* & -\xi_{21} \eta_1^* \ln \eta_1^* \\ \frac{\eta_2^*}{\theta_1^*} \ln \eta_2^* & -\xi_{22} \eta_2^* \ln \eta_2^* \end{vmatrix} = \frac{\eta_1^* \ln \eta_1^* \eta_2^* \ln \eta_2^*}{\theta_1^*} (\xi_{22} - \xi_{21}). \tag{37}$$

Now $-\eta \log \eta$ has its maximum value when $\eta = 1/e$. Also, if the range of permissible values for the times, ξ_1, is sufficiently wide then the expression (37) is maximized by ensuring that $\xi_{22} - \xi_{21}$ shall achieve the maximum possible range $\xi_2(\max) - \xi_2(\min)$ and adjusting the times of reaction so that the anticipated fractions remaining, η_1^*, η_2^*, are each equal to $1/e$. The actual values of ξ_{11} and ξ_{12} required in this case are obtained by substituting $\eta = e^{-1}$ in equation (35). The appropriate times ξ_{11}, ξ_{12}, associated with $\xi_2(\min)$ and $\xi_2(\max)$ are then given by

$$\xi_{11} = \frac{1}{\theta_1^*} \exp\{\theta_2^* \xi_2(\min)\},$$

$$\xi_{12} = \frac{1}{\theta_1^*} \exp\{\theta_2^* \xi_2(\max)\}. \tag{38}$$

It is instructive to consider a numerical example. Suppose the absolute temperature is restricted to the range 380–420° A. It is then convenient to take as an arbitrary base temperature the value $T_0 = 400°$ A. Suppose with time measured in minutes a guessed value for the rate θ_1 at this arbitrary base temperature is $\theta_1^* = 1{\cdot}0$ and a guessed value for $\theta_2^* = 16{,}000$, then we have

$$\theta_1^* = 1{\cdot}0, \quad \theta_2^* = 16{,}000$$

$$380° \text{ A.} \leqslant T = (\xi_2 + 0{\cdot}0025)^{-1} \leqslant 420° \text{ A.}$$

The optimum design will consist of two experiments, the first of which is performed at the minimum temperature of 380° A for time $\xi_{11} = 8{\cdot}21$ min. and the second at the maximum temperature of 420° A for 0·15 min. These times are found by substituting in equation (38) and are simply those values at which, if the guessed values are correct and there were no experimental error, the fraction of product remaining would be $1/e$. The design is illustrated geometrically in Fig. 4. In this figure the design locus in the plane of $\phi_1(\xi_1, \xi_2)$, $\phi_2(\xi_1, \xi_2)$ is bounded by the values $\phi_1(\xi_1, \xi_2) = 0$ and $\phi_1(\xi_1, \xi_2) = -1/\theta_1^* e$, the largest and the smallest values, respectively, that this derivative can take. The straight lines in Fig. 4 are for constant ξ_2 (i.e. for constant temperature) and the curved lines for constant ξ_1 (i.e. for constant time). Where, as we have assumed above, the permissible range of ξ_2 dictates the

design points selected, the design locus is restricted to that portion of the plane within the triangle bounded by the three straight lines

$$\phi_1(\xi_1, \xi_2) = -1/\theta_1^* e,$$
$$\phi_2(\xi_1, \xi_2) = -\theta_1^* \phi_1(\xi_1, \xi_2)\, \xi_2(\text{min}),$$
$$\phi_2(\xi_1, \xi_2) = -\theta_1^* \phi_1(\xi_1, \xi_2)\, \xi_2(\text{max}).$$

The triangle so delineated is itself the triangle of largest area which can be formed with two points on the design locus and the origin with only temperature restricted.

The appropriate design for the case considered is indicated by the two heavy dots in the diagram. It will be seen from the figure that, if ξ_1 were the restricting variable, then a somewhat different result would be obtained, since now the triangle of maximum area would be achieved when the value of $\phi_1(\xi_1, \xi_2)$ fell somewhat short of $-1/\theta_1^* e$.

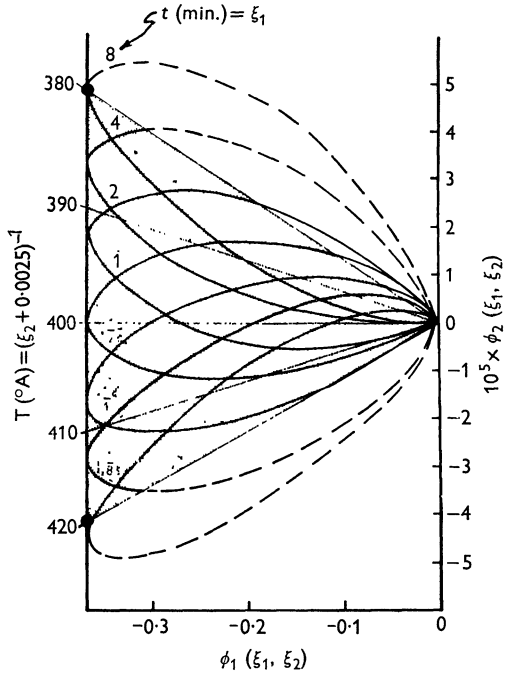

Fig. 4. The design locus for $\eta = \exp\{-1{\cdot}0te^{-16{,}000(T^{-1}-0{\cdot}0025)}\}$. The optimal solution for $T(\text{min}) = 380°\text{A.}$, $T(\text{max}) = 420°$ A. and $0 \leqslant \xi_1 < \infty$ is indicated by heavy dots.

4. Calculation of the adjustments

Suppose a design of this type has been obtained for assumed values of the parameters $\boldsymbol{\theta}^*$, and suppose that the N calculated values at these sets of conditions for $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ are given by the column vector $\boldsymbol{\eta}^*$. Suppose finally that observed responses from the N experiments performed are provided by the column vector $\mathbf{y}$. Then proceeding in the usual way a $p \times 1$ vector of adjustments $\boldsymbol{\delta}$ to be applied to the preliminary values $\boldsymbol{\theta}^*$ are given to a first approximation by

$$\delta = (\mathbf{F}^{*\prime}\mathbf{F}^*)^{-1}\mathbf{F}^{*\prime}(\mathbf{y} - \boldsymbol{\eta}^*)$$

or in the special case where the number of observations N is equal to the number of parameters p by

$$\delta = \mathbf{F}^{*-1}(\mathbf{y} - \boldsymbol{\eta}^*).$$

In the particular example of § 3·3,

$$\mathbf{F}^* = \frac{1}{e}\begin{bmatrix} -1/\theta_1^* & \xi_{21} \\ -1/\theta_1^* & \xi_{22} \end{bmatrix}$$

and

$$\mathbf{y}-\boldsymbol{\eta}^* = \begin{bmatrix} y_2 - e^{-1} \\ y_1 - e^{-1} \end{bmatrix},$$

so that the adjusted values will be

$$\theta_1' = \theta_1^* + \delta_1,$$
$$\theta_2' = \theta_2^* + \delta_2,$$

where δ_1 and δ_2 are the 'regression coefficients' of the elements in $\mathbf{y}-\boldsymbol{\eta}^*$ on the columns of the matrix $\mathbf{F}^*$.

To see the meaning of this suppose that the value of θ_1^* were taken at a base temperature such that $\xi_{21} = -\xi_{22}$, then the adjustments in θ_1^* and θ_2^* are simply

$$\delta_1 = -(\bar{y} - e^{-1})\, e\theta_1^*,$$
$$\delta_2 = (y_1 - y_2)\, e/2\xi_{21}.$$

As would be expected, therefore, when the observed values of y were equal but different from the value e^{-1}, this would imply that adjustment was needed only in the basic rate constant θ_1^*. Unequal values for y_1 and y_2 on the other hand would imply the necessity for adjustment in θ_2^*.

5. Discussion

When parameters are jointly estimated from observations made at particular sets of conditions, it frequently happens that the estimates are highly correlated and consequently have very large variances. This may be simply the result of poor design in which case the first group of estimates may be used as starting values to design a more informative set of trials from which greatly improved estimates can be obtained. Not infrequently, however, it turns out that the unsatisfactory nature of the estimates is a property of the model itself, and that even the best design still yields highly correlated estimates. This point is well illustrated by models such as

$$\eta = \frac{1+\theta_1\xi_1}{\theta_2+\theta_3\xi_1}.$$

In accordance with the method already outlined, the optimal design will be found by choosing three points on the design locus which together with the origin form the tetrahedron of greatest volume. But the design locus for such a model often turns out to be a nearly straight line, so that for any reasonable range of the variable, ξ_1, the volume of the tetrahedron and hence the value of the determinant is small. This arises from the nature of the model itself. Whatever values of the ξ's are chosen in such examples, when one has found one set of values of the θ's that most closely agrees with a given set of data, one can, by making suitable compensating changes in the parameters, obtain other quite different values of the θ's that agree almost as well.

In this connexion it is important to distinguish between the problem we are considering here, which is that of finding designs most efficient for estimating the parameters $\boldsymbol{\theta}$, and a somewhat different problem which is *not* discussed here, that of finding designs such that the response $\eta = f(\boldsymbol{\xi}; \boldsymbol{\theta})$ is determined most precisely in a given region. A particular design may be such that the estimates, $\boldsymbol{\theta}$, of the parameters are highly correlated with large

variances and yet the response, η, is estimated reasonably accurately. The reason for this is not difficult to see. Instability in the $\boldsymbol{\theta}$'s which is *compensating* is by its very nature not such as will cause instability in the estimate of η.

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REFERENCES

Box, G. E. P. (1957). Use of statistical methods in the elucidation of basic mechanisms. *Bull. Inter. Statist. Inst.* **36**, part 3, 215–25.

Brant, J. W. A. (1951). Rate of early growth in domestic fowl. *Poult. Sci.* **30**, 343–61.

Chernoff, H. (1953). Locally optimal designs for estimating parameters. *Ann. Math. Statist.* **24**, 586–602.

Elfving, G. (1952). Optimum allocation in linear regression theory. *Ann. Math. Statist.* **23**, 255–62.

Hendricks, W. A., Jull, M. A. & Titus, H. W. (1932). The utilization of feed by chickens. I. The requirements for growth. *Poult. Sci.* **2**, 74–7.

Koshal, R. S. (1933). Application of the method of maximum likelihood to the improvement of curves fitted by the method of least squares. *J. Roy. Statist. Soc.* **96**, 303–13.

Wald, A. (1943). On the efficient design of statistical investigations. *Ann. Math. Statist.* **14**, 134–40.